

Clinical Risk Assessment Report: Patient 94

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.41x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 2.2%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) Value: -0.19 [Avg (38th %ile)] (-2.2%)

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) Value: -0.16 [Avg (58th %ile)] (-1.2%)

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Patient Age (Age) Value: 67.0 [Avg (64th %ile)] (-0.9%)

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) Value: -0.76 [Bottom 6%] (-0.9%)

Points to continuous, elongated bands of high-density or highly active tumor tissue.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) Value: -0.39 [Avg (28th %ile)] (-0.8%)

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) Value: -0.43 [Bottom <1%] (-0.7%)

Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) Value: -0.1 [Avg (61th %ile)] (-0.6%)

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

PET Radiomic Metabolic Feature (GLZLM_ZLNU.PET) Value: -0.49 [Bottom <1%] (-0.6%)

Metabolic PET feature reflecting glucose avidity and cellular metabolic rate.

Machine Learning Feature Attribution (SHAP Decision Path)

